Experimental verification of the Stefan-Boltzmann law and investigation of the emission and transmission properties of various materials

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The primary objective of this experiment is to verify the Stefan-Boltzmann law and to analyze the thermal emission as well as the infrared transmission properties of various materials. The experiment confirmed the proportionality ($P \propto T^4$) predicted by the Stefan-Boltzmann law for various materials. An estimate for the emissivity factor ϵ was made by comparing the measurement data. Furthermore, the infrared transmittance and dispersion relation were determined using a Leslie cube and a set of filters. In the final part, the Stefan-Boltzmann law was verified for a hollow cylinder at high temperatures. Subsequently, an angledependent measurement at constant temperature demonstrated that the object under investigation is not an ideal Lambertian radiator.

1 Introduction

The study of thermal radiation historically marks the transition from classical thermodynamics to quantum mechanics. Every body with $T > 0K$ holds true emits electromagnetic radiation. The black body plays a central role. This is an idealized body that completely absorbs incident radiation. The emitted power of such a body is described by the Stefan-Boltzmann law [1.

$$P_s = \sigma A T^4 \tag{1}$$

$\sigma = 5.67 * 10^{-8} \frac{W}{m^2 K^4}$ corresponds to the Stefan-Boltzmann constant and A to the surface area of the radiator. In reality, the emitted power is lower than that of an ideal radiator by materialdependent factor $0 < \epsilon < 1$. The actual radiated power is thus given by equation [2].

$$P_{real} = \epsilon \sigma A T^4 \tag{2}$$

This work aims to experimentally investigate the radiation behavior of various surfaces. The goal is to verify the proportionality between radiation power and the fourth power of temperature and to determine the material specific differences. For this purpose, an experimental setup is used that enables the detection of thermal radiation via a thermopile and tests the theoretical predictions of Boltzmann and Kirchhoff.

2 Stefan-Boltzmann law

The experimental confirmation of the radiation laws was crucial for overcoming the "ultraviolet catastrophe" and the development of Planck's quantum hypothesis.

While the Stefan-Boltzmann law [1] provides the theoretical basis for ideal emitters, the Leslie cube allows the investigation of real surfaces under laboratory conditions. This chapter focuses on the comparison between different material properties and their emissivity ϵ. Since radiative power plays a central role in modern sensor technology, the dependence of emission on surface temperature and the exchange with ambient temperature T_0 are practically examined here.

2.1 Materials and methods

To determine the radiant power of different surfaces, a Leslie cube is used. The Leslie cube has a black, a white, a polished nickelplated surface, and a matte nickelplated surface. The water inside the cube is heated in a controlled manner, and the radiant power is measured using a thermopile. The thermopile is located at a distance of 10 cm from the Leslie cube. The experimental setup is shown in Figure [1].

2.2 Results and discussion

In Figure [2], the measured thermoelectric voltage was plotted as a function of the temperature of the Lieslie cube. A graph was created for each face of the cube. To show the T^4 dependence, the measured thermoelectric voltage was plotted as a function of the temperature raised to the power of four. This can be seen in Figure [3]. Since the data points in Figure [3] form a straight line, the T^4 dependence

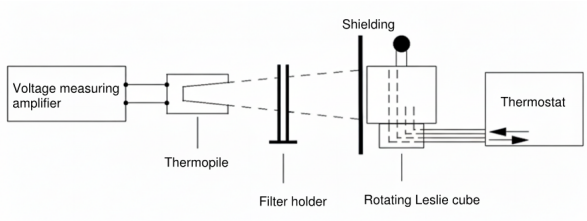

Figure 1: Experimental setup for investigating the emitted radiation power of the four different sides of a Leslie cube. Experimental setup for determining the transmission of infrared radiation through various materials.

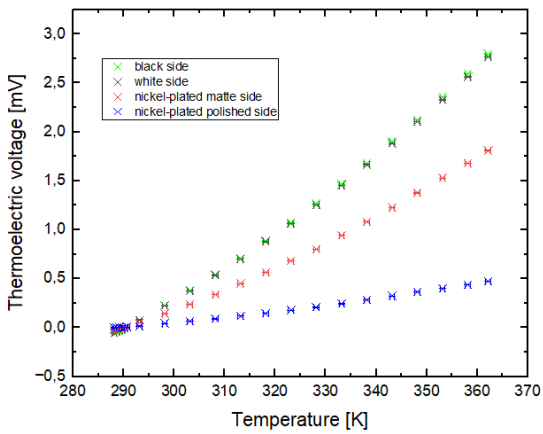

Figure 2: Measured radiation power of the different sides of a Leslie cube as a function of temperature.

is shown. From this experiment, it can be seen that black bodies emit the most radiation, while polished bodies emit the least. This provides insights into the constant ϵ from equation [2]. The blackbody radiator has the highest radiant power and therefore the largest ϵ. The whitebody radiator has almost the same radiant power as the blackbody radiator. This counterintuitive phenomenon occurs because the visual characteristics "white" and "black" only describe the absorption and reflection of light in the visible spectrum. However, at the temperatures of the Leslie cube (approximately 290 K to 370 K), the peak of the thermal emission lies in the infrared spectrum, as dictated by Wien's displacement law. In this infrared spectrum, most nonmetallic paints and coatings have a high emissivity $\epsilon \approx 1$ regardless of their visible color. Consequently, both the visually black and white surfaces behave as nearly ideal thermal emitters in the investigated wavelength range. The constant has the value $\epsilon \approx 1$ for the whitebody. The matte nickelplated surface radiates significantly less than the blackbody radiator, which means $\epsilon < 1$. For the polished nickelplated surface, $\epsilon << 1$. With these ϵ values, one can estimate how much energy these absorb or radiate. The zeros and negative voltage values in figure [2] and [5] can be explained by the operating principle of the thermopile. The operating principle of a thermopile is based on the Seebeck effect. In the Seebeck effect, a voltage is generated between two connected wires made of different materials when a temperature gradient is present. The thermopile thus indicates the temperature difference between a measured object and the surrounding space in the form of a voltage. The thermopile was set to generate a positive voltage when the object being measured is warmer than the room. Therefore, a negative voltage indicates that the room is warmer than the object. The room serves as the reference temperature. When the room and the object being measured are at the same temperature, the voltage is $U = 0.0V$.

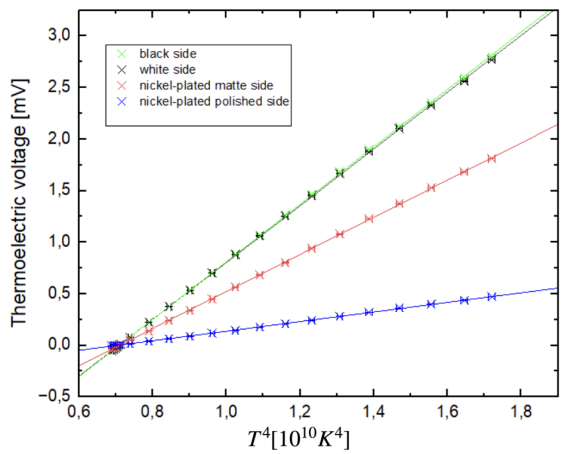

Figure 3: Measured radiation power of the different sides of a Leslie cube as a function of temperature to the power of four.

3 Infrared permeability of various materials

After determining the radiative power of a body, this experiment deals with the transmission of infrared radiation through various materials. Using Wien's displacement law [3], one can determine the wavelength at which a body emits the most radiation at a given temperature.

$$\lambda_{max} = \frac{2898\mu m * K}{T} \tag{3}$$

If the maximum lies in the infrared range, the absorption and transmission properties of a material for infrared radiation can be determined. Determining the absorption properties of a material gives important information about its effects as well as for certain environmental phenomena such as the greenhouse effect.

3.1 Material and methods

The experimental setup is shown in Figure [1]. In contrast to the experiment [2], different materials are used in the filter holder. The materials window glass, silicon, infrared filter and singlecrystal NaCl are investigated. The black surface of the Leslie cube continuously shines onto the different filters.

3.2 Results and discussion

At a temperature of $T = 348.15 \pm 0.50\,\mathrm{K}$, the equation [3] yields a wavelength maximum of $\lambda_{max} = 8.324 \pm 0.012\,\mu m$ which is in the infrared range. First, the radiant power of the black side is measured. This value serves as a reference to determine the transmitted radiation relative to the total radiation. Table [1] lists the measured radiant power and its ratio to the total radiation.

Material	Value $[mV]$	Transmission [%]
Reference value	2.114	-
Window glass	0.051	2.412 ± 0.047
Silicon	0.974	46.074 ± 0.052
Infrared filter	0.763	36.093 ± 0.050
NaCl	1.587	75.071 ± 0.059

Table 1: Measured radiant power, which indicates the transmission of different materials in percent relative to the reference value.

The error in the measured radiant power is $\sigma = \pm 0.001 mV$ and is the same for all measurements. The measurements show that the transmission coefficient is wavelengthdependent. While visible light is transmitted by window glass almost without loss, infrared radiation is largely absorbed. While visible light is transmitted by window glass almost without a loss, infrared radiation is largely absorbed. For NaCl visible light is completely reflected and absorbed, while approximately 75% of the infrared radiation is transmitted.
The measured transmission coefficients align well with standard literature values for the mid infrared spectrum (around $8\,\mu$m). As expected, standard window glass is practically opaque (not letting light pass through) in this wavelength range, which is consistent with the measured transmission of only 2.4%. Uncoated silicon typically exhibits a theoretical transmission limit of approximately 50% due to high Fresnel reflection losses caused by its large refractive index. The experimental result of 46.1% is in a good range with this theoretical limit. Single crystal NaCl is known for its high broadband infrared transparency, typically exceeding 90%. The slightly lower measured value of 75.1% can likely be attributed to scattering and absorption losses caused by surface degradation. NaCl is highly hygroscopic (attracting and absorbing water vapor), which degrades the optical quality of the crystal surfaces over time.
The wavelength emitted most at a room temperature of $T_R = 290.15K$ is $\lambda_{R,max} = 9.988\mu m$. This is also in the infrared spectral range. Since the thermopile measures a temperature difference, background radiation is not included in the measurement result.

4 Blackbody radiation

A blackbody behaves as described in [2]. The radiance of the radiation emitted by the blackbody is angleindependent. However, the measured radiant intensity J is angledependent and follows Lambert's law [4].

$$J = J_0 * cos(\theta) \tag{4}$$

After changing the angular measurement position θ the observed solid angle Ω also changes and the measured radiation intensity decreases. A body that radiates in this way is called a Lambertian radiator. The Lambertian characteristic is the goal of many technologies in industry. A smartphone screen should be a Lambertian radiator so that the display appears equally bright from any position. Diffusers in front of lamps are also designed to enforce a Lambertian distribution, thus avoiding harsh shadows and glare.

4.1 Material and methods

The blackbody radiator consists of an electric oven that heats a hollow cylinder. The oven temperature is adjustable and measured directly. The radiation emitted by the blackbody radiator is measured using a thermopile. The thermopile can be positioned at an angle from $\theta_{min} = 0°$ to $\theta_{max} = 50°$ at a constant distance in front of the radiator. Watercooled baffles are mounted between the blackbody radiator and the thermopile. These baffles ensure that the thermopile measures only the radiation from the cavity and does not also detect radiation from the walls of the hot oven.

4.2 Results and discussion

At an angle of $\theta = 0°$, the thermoelectric voltage across the thermopile is measured over a temperature range from $T_{min} = 293.15K$ to $T_{max} = 673.15K$. The results are shown in Figures [4] and [5].
Since the thermoelectric voltage has a linear relationship with T^4, the radiant power of the heated hollow cylinder also follows the Stefan-Boltzmann law [1]. To verify Lambert's law [4], the body was heated to a constant temperature of $T = 623.15K$. The thermopile was positioned in $\Delta\theta = 5°$ increments from $\theta_{min} = 0°$ to $\theta_{max} = 50°$, and the measured

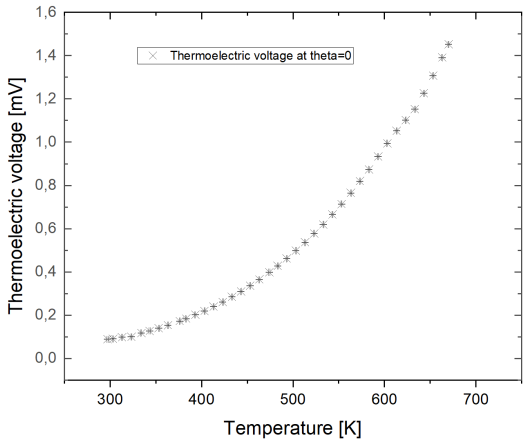

Figure 4: Measured radiant power of a heated black hollow cylinder at different temperatures. The measurement was performed at an angle of $\theta = 0°$.

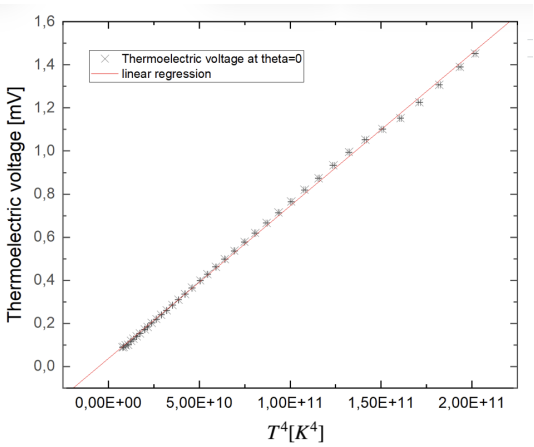

Figure 5: Measured radiant power of a heated black hollow cylinder at different temperatures. The data are plotted against temperature to the power of four. The measurement was performed at an angle of $\theta = 0°$

values were recorded. θ_{min} corresponds to the position directly in front of the opening of the hollow cylinder. The measured values are shown in Figures [6] and [7].

The measured data do not perfectly follow the theoretical curve, as shown in Figure [6]. Nevertheless, a regression line can be constructed from the measurement points, as shown in Figure [7]. The deviations can be explained by systematic errors, such as the oven's temperature setting. Since the oven is controlled directly by a voltage and not by a thermostat, temperature fluctuations occurred during the measurement. This error can explain the deviations from the theory. It can be seen that the heated hollow cylinder resembles a Lambertian radiator.

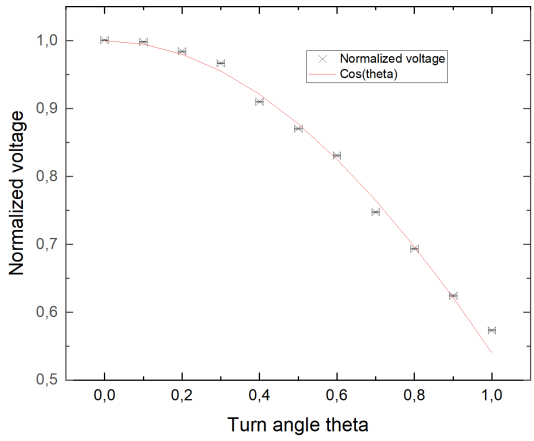

Figure 6: Measured radiation power at constant temperature $T = 623.15K$ at different angles compared to the theoretical curve.

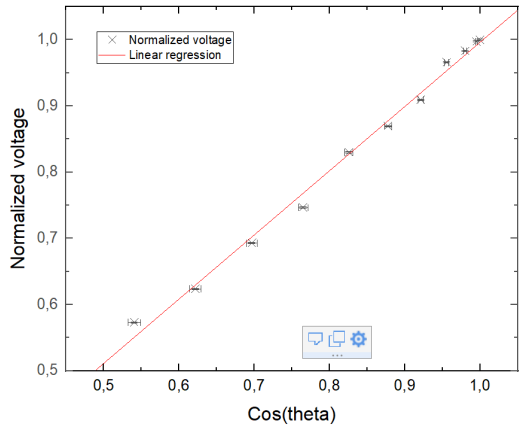

Figure 7: Measured radiant power at constant temperature $T = 623.15K$ at various angles. The values are plotted against the cosine of the angle.

5 Conclusion

In this series of experiments, the emissivity of real materials was investigated by using a Leslie cube. For this purpose, the radiant power was measured at different temperatures using a thermopile. A linear relationship with T^4 was demonstrated for a black, white, nickelplated matte, and nickelplated polished face of the cube. This corresponds to the predictions of the Stefan-Boltzmann law. By comparing the measured stresses, the material dependent emissivity factor could be estimated. The black and white faces of the cube have a factor $\epsilon \approx 1$, where $\epsilon_b > \epsilon_w$. For the matte face, $\epsilon_m < 1$, and the polished face has $\epsilon << 1$.
The infrared transmittance of various materials was then determined. For this purpose, a filter was inserted between the Leslie cube and the thermopile. The measured voltage with and without the filter were compared to determine the percentage of trans-

mitted infrared radiation. A dispersion relationship for the transmittance factor was established.
In the last part, a hollow cylinder was heated by an electric oven. The emitted radiation was again recorded using a thermopile, and a linear relationship with T^4 was observed. To test whether it was a Lambertian radiator, the radiant power was measured at a fixed temperature at various angles. Small deviations from the theoretically determined curve were found. These could be explained by systematic errors. The conclusion was that it was a radiator that closely resembled a Lambertian radiator.

References

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